

Vitalis Clinic

A multidisciplinary clinic management platform — four roles, four core functions, and a business logic designed to be adaptable for real-world deployment.

 Booking · Medical Records · Payments · Notifications

 FR / EN

 GDPR by design

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1 SCOPE

What Vitalis Clinic is

Vitalis Clinic is a vertical business product delivered per instance: each buyer gets their own deployment, adaptable to their market (payment method, language, notification channels). The goal isn't an isolated technical demo, but proof that a complex, credible business product can be steered end to end — from business logic through to production.

Patient

Books, reschedules, manages their account and data.

Doctor

Real-time schedule, medical record, leave requests.

Front desk

Human follow-up, alert handling, walk-ins.

Management

Audit, reporting, leave approval.

Key design principle — nothing that showcases the build itself (presentation deck, tech stack, performance score) appears on the product. The product must stay credible as a real clinic site; its showcase lives elsewhere, in a separate portfolio.

How the project was steered

A complex business product, like a hospital-grade software system, tests the ability to frame a problem before building it – not just to write code. The project rests on three pillars.

1 Business framing Every rule (payment, GDPR, reminder cascade) is specified, then challenged through a critical review to surface contradictions before any code is written.	2 AI-assisted build Code is produced with an AI agent, in short cycles per functional phase – never a single monolithic pass.	3 Multi-layer verification Automated static analysis, review in a separate context for each phase, and a full review before any public launch.
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💡 A note on method: the business logic in this very document was itself put through the same scrutiny to eliminate contradictions – two phrasings of the same rule that didn't line up, a role able to trigger an action it was never authorized to perform, a core table missing for a function announced as a top priority. Framing a project well also means knowing how to challenge it before building it.

Build sequence

The steps follow the product's real dependencies — you can't handle a payment before an appointment exists, or a reminder cascade before payments are reliable.

-  **1 · Foundations**
Authentication, roles, real-time availability calculation, simple booking.
 -  **2 · Account & compliance**
Account management, GDPR, demo mode, multilingual support, time zones.
 -  **3 · Rescheduling & cancellation**
Rescheduling logic, doctor cancellation, schedule exceptions.
 -  **4 · Payments**
Stripe + PayPal, statuses, security audit of the inbound webhook.
 -  **5 · Notifications**
Reminder cascade, staff alerts, follow-up loop — the densest part of the project.
 -  **6 · Billing**
Automatic invoicing, credit notes on administrative correction.
 -  **7 · Doctor leave**
Request, management approval, conflicts.
 -  **8 · Hardening**
Automated tests, real concurrency, tested backups — the last step before any presentation.
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Account & compliance

Stable identity

The patient identifier (UUID) never changes, regardless of what's edited on the account. Fields with no security stake (name, phone) can be edited in place; sensitive fields (email, password) live in a dedicated section with double confirmation and session invalidation.

GDPR

Data export and deletion are self-service. Deletion doesn't erase the account: it anonymizes it (the UUID is kept to preserve referential integrity with past appointments and payments), while login access is genuinely deleted. Future appointments are cancelled silently — no staff alert, since there's nothing to fix — whereas a missing profile caused by a technical anomaly does trigger an alert: the difference between an intended consequence and a real problem is encoded in the data itself.



Multilingual

FR/EN from launch, architected so a new language is an addition, not a rebuild.



Time zones

Storage and automation run in UTC; every user sees times converted to their own local zone, never the other party's.

Booking, availability & rescheduling

A slot's availability is never a field to keep in sync: it's recalculated live from the recurring schedule, one-off exceptions, and active appointments. Patient and doctor always read the same truth. The same time slot can never be booked twice, even by two patients acting at the exact same instant — that guarantee is enforced at the database level, not just in the interface.

Golden rule of rescheduling — rescheduling never modifies an existing appointment: it always creates a new record, linked back to the original for traceability. Any payment already made automatically follows the new date, with no decision required from the patient. The whole operation (cancel + create + move the payment) succeeds or fails as a single unit — never an inconsistent in-between state.

A patient can reschedule freely up to 3 hours before their appointment; below that, they go through the front desk. This threshold only protects upcoming appointments — it doesn't apply to the 15-day window granted after a no-show, which follows a different logic (see next section).

Payments & edge cases

Stripe is the primary payment method, PayPal a secondary option tested to the same standard. Payment happens at the time of booking; the slot is held during the transaction, then automatically released if payment doesn't go through within a short window — no appointment ever persists without a confirmed payment.

The 15-day rule

A no-show doesn't refund the payment: it puts it on hold for 15 days, during which the patient can reschedule using that same payment. Past that window without rescheduling, the payment becomes permanently earned by the clinic. Reporting never counts a payment on hold as earned revenue — only final statuses feed into management's dashboard.

 The same payment can never be used twice: if the rescheduled appointment is also missed, the payment immediately becomes final, with no second 15-day window opening up.

7 BUSINESS LOGIC

Reminder cascade & staff alerts

Goal: reduce missed appointments without harassing the patient. The patient picks one, two, or three channels (WhatsApp, Telegram, Email) in a fixed priority order. Before every send, the appointment's status is re-read live — if the patient has already responded, the cascade stops immediately, regardless of the remaining channels.

Step	Delay	Condition
Priority channel	T+0	Checked and status still pending
Next channel	+5h	Checked and status still pending
Last channel	+5h	Checked and status still pending

Hard circuit breaker: within 3 hours of the appointment, the cascade stops, no matter how far along it is. No automated sends between 9pm and 7am (patient's local time), unless delaying would miss the reminder's useful window entirely.

"Needs attention" queue — eight distinct situations (exhausted cascade, total technical failure, doctor cancellation, schedule conflict...) converge into a single staff alert mechanism, with a label derived from the appointment's state rather than a dedicated field per case. A bounded human follow-up loop, capped at two call cycles, takes over when manual intervention is needed.

Doctor cancellations & leave

A doctor can cancel an individual appointment at any time, including within 3 hours, with no justification required — a genuine unavailability shouldn't need to be explained. This case automatically locks the slot (it never reopens on its own), unlike a patient-initiated cancellation, where the slot becomes available again normally. This single distinction lets the rest of the system know, unambiguously, who initiated a cancellation.

Full leave (a day off or an extended absence) follows a different path than a simple schedule exception: it goes through a request submitted for management approval, whose sign-off also counts as confirmation of any conflicting cancellations — no redundant extra step.

9 BUSINESS LOGIC

Billing & medical record



Billing

An invoice is generated once, when an appointment is marked completed. A later administrative correction (data entry error) automatically generates a credit note — the original invoice stays on record, never deleted.



Medical record

One per patient, updated at every consultation. Read access for any doctor treating that patient (continuity of care), write access limited to the author. Anonymized, never deleted, on GDPR deletion.

10 GOVERNANCE

Roles & permissions

Role	Can	Cannot
Patient	Book, reschedule, manage their account, export/delete their data	Cancel online within 3h, see other patients
Doctor	Real-time schedule, free individual cancellation, medical records of their own patients	Write to a patient's identity fields, take full leave without approval
Front desk	Handle alerts, reschedule manually, follow up, issue tracked refunds	Access the medical record or the audit log
Management	Approve leave, view the audit log and reporting, full read access	Edit a patient's data, act directly on an individual appointment

The audit log — record access and GDPR events — is append-only and restricted to management: no line is ever edited or deleted, by any role.

What this project demonstrates

Vitalis Clinic isn't a technical showpiece: it's a test of the ability to think a business product through to its edge cases, then build it in a steered way rather than a reactive one.



Structure before building

Four roles, four core functions, explicit dependencies between steps — the product is designed as a system before it's written as code.



Challenging your own logic

Every business rule was re-read for its own contradictions before being validated — a discipline as important as writing the code itself.



Steering AI agents

Framing, building, verifying: three distinct roles, deliberately kept separate rather than collapsed into a single act of production.



Thinking business, not just tech

GDPR, time zones, no-shows, refunds: the real difficulties of a clinical product are anticipated, not discovered in production.